

# ELITE IGCSE ACADEMY

*Student Past Paper Solutions*

## Nov 2023 P1H Worked Solutions

Nov 2023 | P1H

29 questions   ■   question image followed by worked solution

PREPARED BY

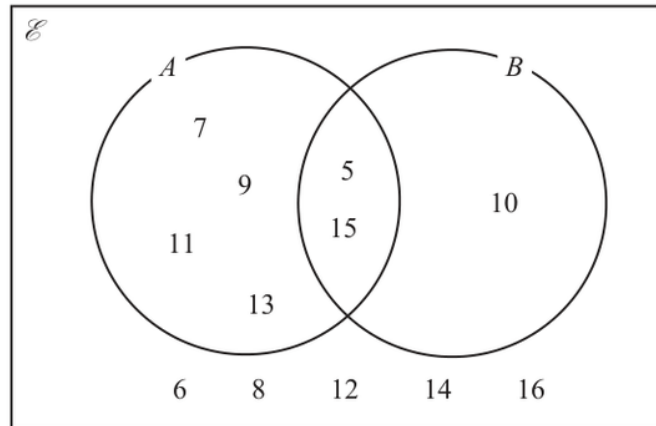
**Dr. Eslam Ahmed**

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**PAST PAPER QUESTION****Q#: 1****Marks: 3****TOPIC: Set Notation & Venn Diagrams**

Nov 2023 P1H - Nov 2023 | P1H

**1** Here is a Venn diagram.

List the members of the set

(a)  $A$ .....  
(1)(b)  $A \cap B$ .....  
(1)(c)  $(A \cup B)'$ .....  
(1)**(Total for Question 1 is 3 marks)**

## PAST PAPER SOLUTION

Q#: 1

Marks: 3

TOPIC: **Set Notation & Venn Diagrams**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 2****Marks: 6****TOPIC: Forming & Solving Equations**

Nov 2023 P1H - Nov 2023 | P1H

- 2 (a) Factorise fully  $12pq - 18p$

.....  
(2)

There are 56 metal bars in a box.  
Each metal bar is gold or silver or zinc.

$y$  metal bars are gold.

$(3y + 7)$  metal bars are silver.

$(2y - 5)$  metal bars are zinc.

- (b) Work out the number of metal bars that are zinc.  
Show clear algebraic working.

.....  
(4)

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(Total for Question 2 is 6 marks)

## PAST PAPER SOLUTION

Q#: 2

Marks: 6

TOPIC: **Forming & Solving Equations**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 3****Marks: 4**TOPIC: **Percentages**

Nov 2023 P1H - Nov 2023 | P1H

**3** Joshua buys a car for \$12 500

He sells the car to Nina.

Nina pays

- a deposit of \$1500
- followed by 36 monthly payments of \$450

Work out Joshua's percentage profit.

..... %

**(Total for Question 3 is 4 marks)**

EA

## PAST PAPER SOLUTION

Q#: 3

Marks: 4

TOPIC: Percentages

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 4****Marks: 4****TOPIC: Probability Toolkit**

Nov 2023 P1H - Nov 2023 | P1H

- 4 A biased spinner has three sections each of a different colour.

The table shows the probability that, when the spinner is spun once, it will land on blue or on orange or on white.

| Colour      | blue | orange | white |
|-------------|------|--------|-------|
| Probability | 0.58 | $2x$   | $x$   |

- (a) Work out the value of  $x$

$$x = \dots\dots\dots (2)$$

The spinner is spun 250 times.

- (b) Work out an estimate for the number of times the spinner will land on blue.

$$\dots\dots\dots (2)$$

**(Total for Question 4 is 4 marks)**

## PAST PAPER SOLUTION

Q#: 4

Marks: 4

TOPIC: **Probability Toolkit**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 5****Marks: 3****TOPIC: Circles, Arcs & Sectors**

Nov 2023 P1H - Nov 2023 | P1H

- 5 The diagram shows a shaded shape made from three identical semicircles,  $X$ ,  $Y$  and  $Z$

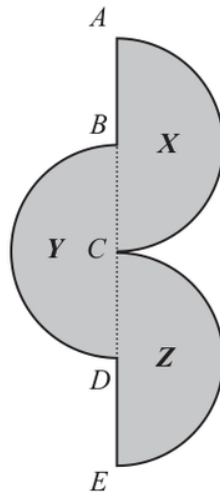


Diagram **NOT**  
accurately drawn

$ABCDE$  is a straight line.

$AC$  is the diameter of semicircle  $X$  and  $B$  is the centre of semicircle  $X$

$BD$  is the diameter of semicircle  $Y$  and  $C$  is the centre of semicircle  $Y$

$CE$  is the diameter of semicircle  $Z$  and  $D$  is the centre of semicircle  $Z$

$$AC = BD = CE = 20 \text{ cm}$$

Work out the perimeter of the shaded shape.

Give your answer correct to the nearest whole number.

..... cm

(Total for Question 5 is 3 marks)

## PAST PAPER SOLUTION

Q#: 5

Marks: 3

TOPIC: **Circles, Arcs & Sectors**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 6****Marks: 4****TOPIC: Percentages**

Nov 2023 P1H - Nov 2023 | P1H

- 6 Juan wants to buy a ticket to fly from Madrid to Berlin.

He finds two different types of ticket he can buy in a sale, ticket **A** and ticket **B**

ticket **A**  
 $\frac{1}{6}$  off normal price

ticket **B**  
20% off normal price

The sale price of ticket **A** is 140 euros.

The sale price of ticket **B** is 136 euros.

Work out the difference between the normal price of ticket **A** and the normal price of ticket **B**

..... euros

(Total for Question 6 is 4 marks)

## PAST PAPER SOLUTION

Q#: 6

Marks: 4

TOPIC: Percentages

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 7****Marks: 2****TOPIC: Prime Factors, HCF & LCM**

Nov 2023 P1H - Nov 2023 | P1H

7  $A = 5^3 \times 7^3 \times 11^6$  and  $B = 5^6 \times 7^2 \times 11^4$

Find the highest common factor (HCF) of  $A$  and  $B$

Give your answer as a product of powers of its prime factors.

(Total for Question 7 is 2 marks)

EA

## PAST PAPER SOLUTION

Q#: 7

Marks: 2

TOPIC: **Prime Factors, HCF & LCM**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

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No worked solution is saved yet.

EA

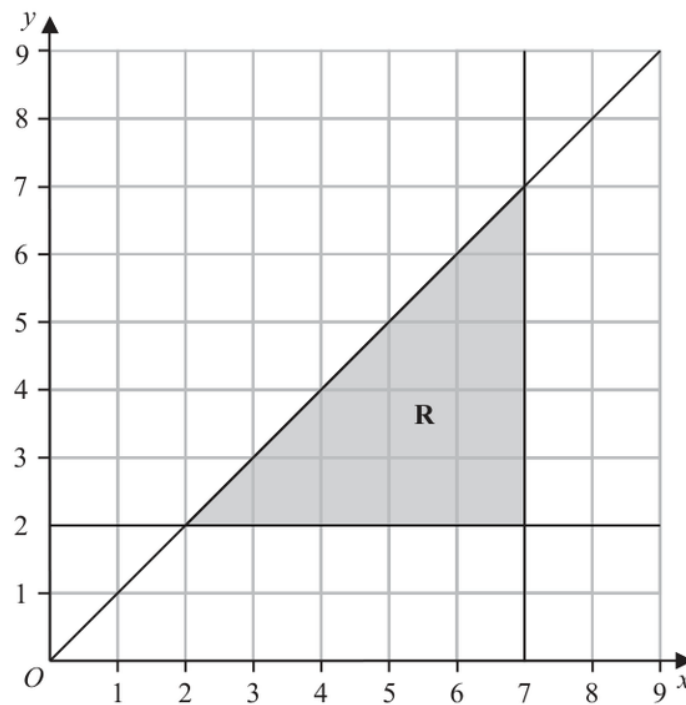
**PAST PAPER QUESTION****Q#: 8****Marks: 5****TOPIC: Graphing Inequalities**

Nov 2023 P1H - Nov 2023 | P1H

- 8 (a) Solve the inequality  $8x - 4 \geq 3x - 10$

.....  
(2)

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



- (b) Write down the three inequalities that define the region **R**

.....  
(3)

(Total for Question 8 is 5 marks)

## PAST PAPER SOLUTION

Q#: 8

Marks: 5

TOPIC: **Graphing Inequalities**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 9****Marks: 4****TOPIC: Powers, Roots & Standard Form**

Nov 2023 P1H - Nov 2023 | P1H

- 9 (a) Write  $5.87 \times 10^{-4}$  as an ordinary number.

.....  
(1)

- (b) Write 84 000 000 in standard form.

.....  
(1)

The number of neurons in a human brain is  $8.5 \times 10^{10}$   
 The number of neurons in a monkey brain is  $1.47 \times 10^9$

The number of neurons in a human brain is  $K \times$  the number of neurons in a monkey brain.

- (c) Work out the value of  $K$   
 Give your answer correct to one decimal place.

$K =$  .....  
(2)

(Total for Question 9 is 4 marks)

## PAST PAPER SOLUTION

Q#: 9

Marks: 4

TOPIC: **Powers, Roots & Standard Form**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

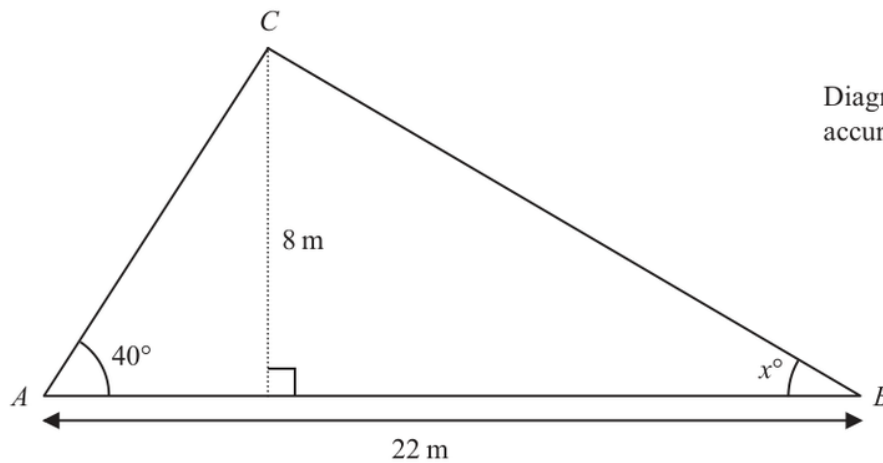
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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 10****Marks: 5****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2023 P1H - Nov 2023 | P1H

**10** Here is triangle  $ABC$ 

Work out the value of  $x$   
Give your answer correct to one decimal place.  
Show your working clearly.

 $x = \dots\dots\dots$ **(Total for Question 10 is 5 marks)**

## PAST PAPER SOLUTION

Q#: 10

Marks: 5

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 11****Marks: 3****TOPIC: Algebraic Fractions**

Nov 2023 P1H - Nov 2023 | P1H

**11** Express  $\frac{3}{4} + \frac{5-x}{6x}$  as a single fraction in its simplest terms.

(Total for Question 11 is 3 marks)

EA

## PAST PAPER SOLUTION

Q#: 11

Marks: 3

TOPIC: **Algebraic Fractions**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

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No worked solution is saved yet.

EA

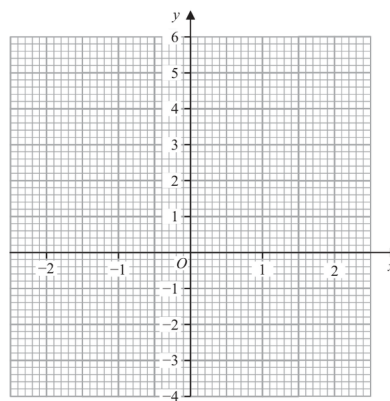
**PAST PAPER QUESTION****Q#: 12****Marks: 5****TOPIC: Graphs of Functions**

Nov 2023 P1H - Nov 2023 | P1H

**12** (a) Complete the table of values for  $y = x^3 - 3x + 1$ 

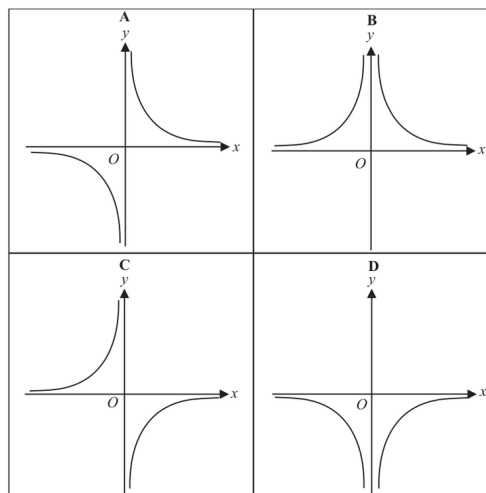
|     |    |    |   |    |   |
|-----|----|----|---|----|---|
| $x$ | -2 | -1 | 0 | 1  | 2 |
| $y$ |    |    |   | -1 |   |

(2)

(b) On the grid, draw the graph of  $y = x^3 - 3x + 1$  for values of  $x$  from -2 to 2

(2)

Here are four graphs.

(c) Write down the letter of the graph that could have the equation  $y = -\frac{1}{x^2}$ 

(1)

**(Total for Question 12 is 5 marks)**

## PAST PAPER SOLUTION

Q#: 12

Marks: 5

TOPIC: **Graphs of Functions**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

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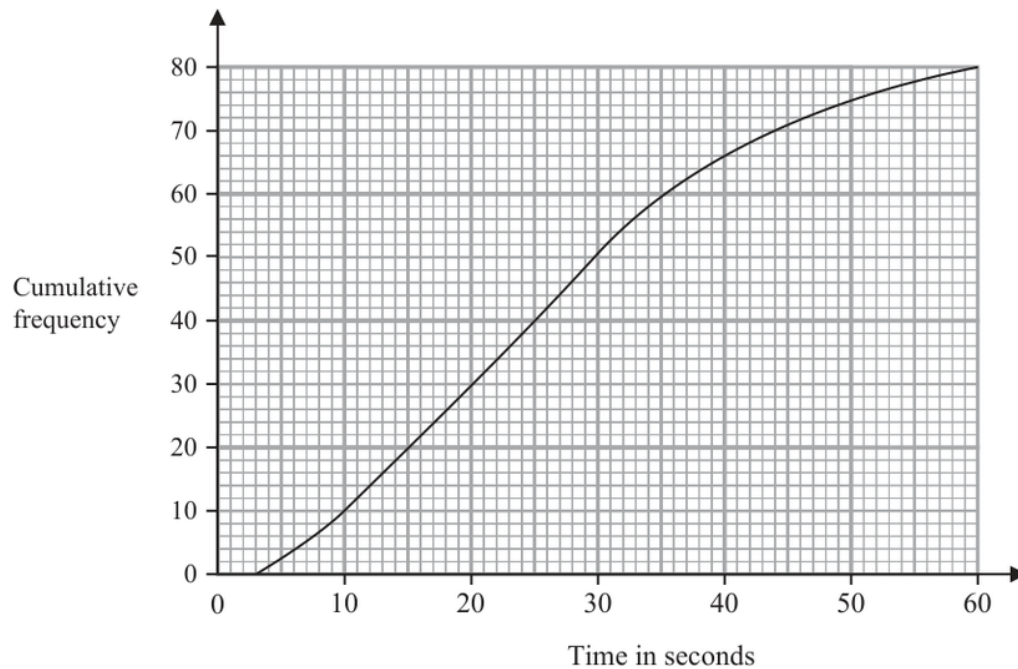
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EA

**PAST PAPER QUESTION****Q#: 13****Marks: 4****TOPIC: Cumulative Frequency Diagrams**

Nov 2023 P1H - Nov 2023 | P1H

- 13** The cumulative frequency graph gives information about the times, in seconds, that 80 adults took to log in to an online bank.



- (a) Find an estimate for the median time.

..... seconds  
(1)

- (b) Work out the percentage of these adults that took longer than 50 seconds to log in.  
Show your working clearly.

..... %  
(3)

(Total for Question 13 is 4 marks)

## PAST PAPER SOLUTION

Q#: 13

Marks: 4

TOPIC: **Cumulative Frequency Diagrams**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

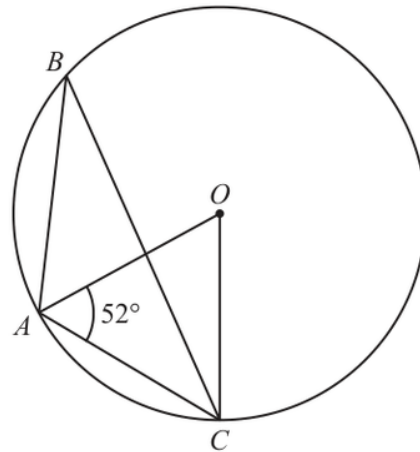
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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 14****Marks: 3****TOPIC: Circle Theorems**

Nov 2023 P1H - Nov 2023 | P1H

**14**Diagram **NOT**  
accurately drawn $A$ ,  $B$  and  $C$  are points on a circle, centre  $O$ Angle  $OAC = 52^\circ$ Find the size of angle  $ABC$ 

Give reasons for your working.

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**(Total for Question 14 is 3 marks)**

## PAST PAPER SOLUTION

Q#: 14

Marks: 3

TOPIC: **Circle Theorems**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 15****Marks: 3****TOPIC: Rearranging Formulas**

Nov 2023 P1H - Nov 2023 | P1H

**15** Make  $n$  the subject of the formula  $x = \frac{3p+n}{3n-4}$

(Total for Question 15 is 3 marks)

EA

## PAST PAPER SOLUTION

Q#: 15

Marks: 3

TOPIC: **Rearranging Formulas**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 16****Marks: 4****TOPIC: Differentiation**

Nov 2023 P1H - Nov 2023 | P1H

**16** A curve has equation  $y = 4x^3 - 8x + 5$

Find the  $x$  coordinates of the two points on the curve where the gradient is  $\frac{1}{3}$

**(Total for Question 16 is 4 marks)**

EA

## PAST PAPER SOLUTION

Q#: 16

Marks: 4

TOPIC: **Differentiation**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

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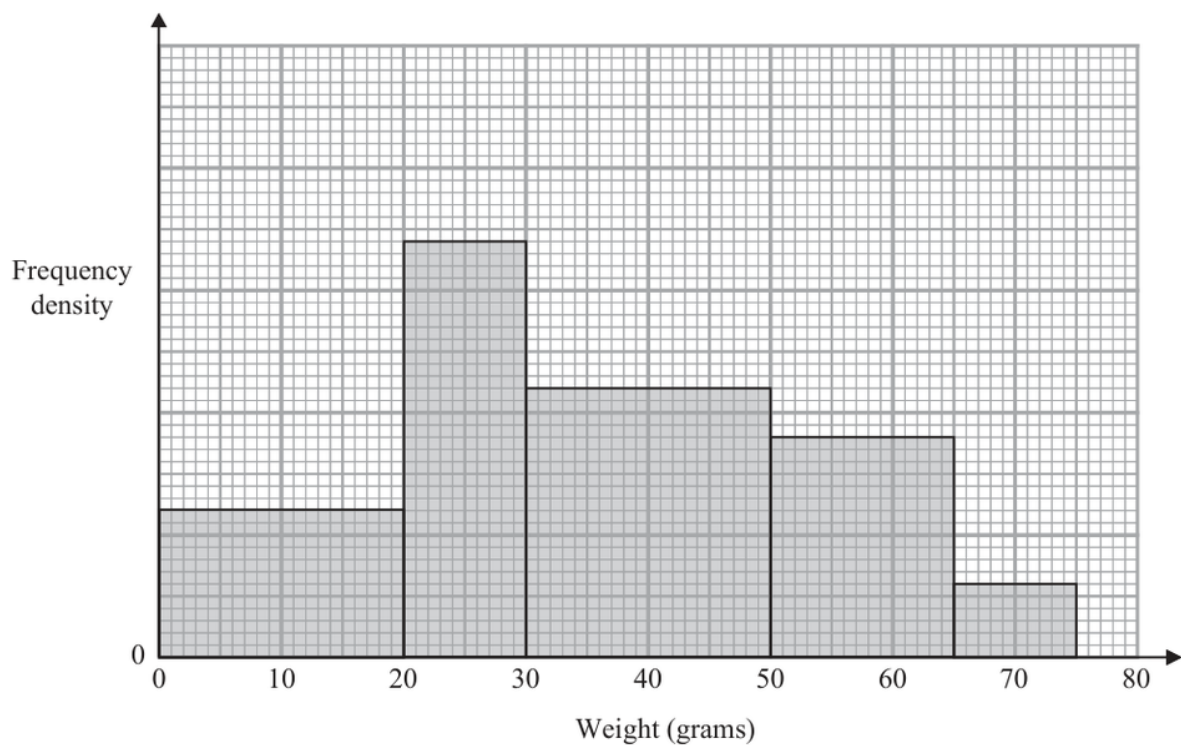
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EA

**PAST PAPER QUESTION****Q#: 17****Marks: 3****TOPIC: Histograms**

Nov 2023 P1H - Nov 2023 | P1H

17 The histogram gives information about the weights, in grams, of some oranges in a box.



24 of these oranges weigh less than 20 grams.

Medium oranges weigh between 35 grams and 55 grams.

Work out an estimate for the number of medium oranges in the box.

**(Total for Question 17 is 3 marks)**

## PAST PAPER SOLUTION

Q#: 17

Marks: 3

TOPIC: **Histograms**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

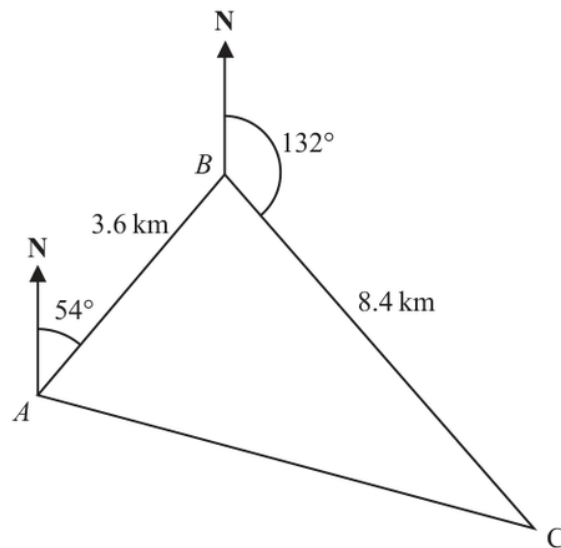
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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 18****Marks: 5****TOPIC: Bearings, Scale Drawing & Constructions**

Nov 2023 P1H - Nov 2023 | P1H

**18** The diagram shows the positions of three villages,  $A$ ,  $B$  and  $C$ Diagram **NOT**  
accurately drawnThe bearing of  $B$  from  $A$  is  $054^\circ$ The bearing of  $C$  from  $B$  is  $132^\circ$ Melur walks from  $A$  to  $B$ She then walks from  $B$  to  $C$  and from  $C$  to  $A$ Melur walks at an average speed of  $6 \text{ km/h}$ 

Work out the total time Melur takes.

Give your answer in hours and minutes.

..... hours ..... minutes

(Total for Question 18 is 5 marks)

## PAST PAPER SOLUTION

Q#: 18

Marks: 5

TOPIC: **Bearings, Scale Drawing & Constructions**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 19****Marks: 6**TOPIC: **Sequences**

Nov 2023 P1H - Nov 2023 | P1H

**19** Here are the first 4 terms in an arithmetic sequence.

3      7      11      15

The last term of the sequence is  $x$

The sum of the terms of the sequence is 7260

Find the value of  $x$

Show clear algebraic working.

$x =$  .....

(Total for Question 19 is 6 marks)

EA

## PAST PAPER SOLUTION

Q#: 19

Marks: 6

TOPIC: **Sequences**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 20****Marks: 4****TOPIC: Probability Toolkit**

Nov 2023 P1H - Nov 2023 | P1H

**20** A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag.  
He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins.

The probability that the two coins have a mean value of 10 cents is  $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

---

**(Total for Question 20 is 4 marks)**

EA

## PAST PAPER SOLUTION

Q#: 20

Marks: 4

TOPIC: **Probability Toolkit**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 20****Marks: 4****TOPIC: Probability Toolkit**

Nov 2023 P1H - Nov 2023 | P1H

**20** A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag.  
He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins.

The probability that the two coins have a mean value of 10 cents is  $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

---

**(Total for Question 20 is 4 marks)**

EA

## PAST PAPER SOLUTION

Q#: 20

Marks: 4

TOPIC: **Probability Toolkit**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

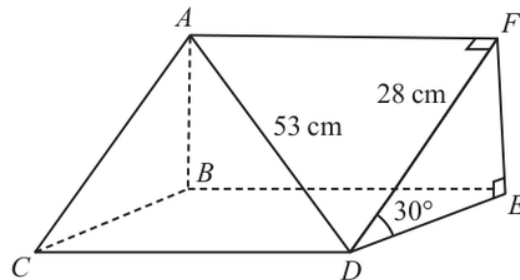
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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 21****Marks: 5****TOPIC: Volume & Surface Area**

Nov 2023 P1H - Nov 2023 | P1H

**21** Here is a triangular prism  $ABCDEF$ Diagram **NOT**  
accurately drawn

$$AD = 53 \text{ cm}$$

$$DF = 28 \text{ cm}$$

$$\text{Angle } FDE = 30^\circ$$

Work out the volume of the triangular prism.

Give your answer correct to the nearest whole number.

.....  $\text{cm}^3$ 

(Total for Question 21 is 5 marks)

## PAST PAPER SOLUTION

Q#: 21

Marks: 5

TOPIC: **Volume & Surface Area**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

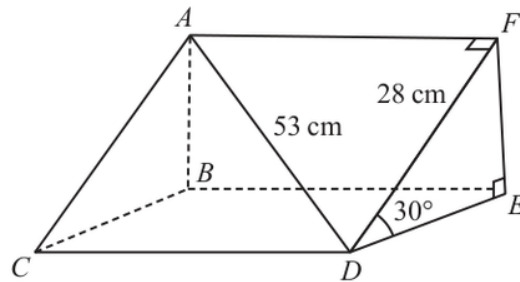
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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 21****Marks: 5****TOPIC: Volume & Surface Area**

Nov 2023 P1H - Nov 2023 | P1H

**21** Here is a triangular prism  $ABCDEF$ Diagram **NOT**  
accurately drawn

$$AD = 53 \text{ cm}$$

$$DF = 28 \text{ cm}$$

$$\text{Angle } FDE = 30^\circ$$

Work out the volume of the triangular prism.

Give your answer correct to the nearest whole number.

.....  $\text{cm}^3$ 

(Total for Question 21 is 5 marks)

## PAST PAPER SOLUTION

Q#: 21

Marks: 5

TOPIC: **Volume & Surface Area**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 22****Marks: 6****TOPIC: Coordinate Geometry**

Nov 2023 P1H - Nov 2023 | P1H

- 22** [In this question 1 cm = 1 unit on the  $x$ -axis and  
1 cm = 1 unit on the  $y$ -axis]

$P$  is a point on a circle with centre  $(0, 0)$

The coordinates of  $P$  are  $(8, -10)$

The line  $L$  is the tangent to the circle at the point  $P$

$L$  crosses the  $x$ -axis at the point  $Q$  and crosses the  $y$ -axis at the point  $R$

Work out the length of  $RQ$

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 22 is 6 marks)

EA

## PAST PAPER SOLUTION

Q#: 22

Marks: 6

TOPIC: **Coordinate Geometry**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 22****Marks: 6****TOPIC: Coordinate Geometry**

Nov 2023 P1H - Nov 2023 | P1H

- 22** [In this question 1 cm = 1 unit on the  $x$ -axis and  
1 cm = 1 unit on the  $y$ -axis]

$P$  is a point on a circle with centre  $(0, 0)$

The coordinates of  $P$  are  $(8, -10)$

The line  $L$  is the tangent to the circle at the point  $P$

$L$  crosses the  $x$ -axis at the point  $Q$  and crosses the  $y$ -axis at the point  $R$

Work out the length of  $RQ$

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 22 is 6 marks)

EA

## PAST PAPER SOLUTION

Q#: 22

Marks: 6

TOPIC: **Coordinate Geometry**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION

Q#: 23

Marks: 5

TOPIC: Area & Volume of Similar Shapes

Nov 2023 P1H - Nov 2023 | P1H

23 Solid A is similar to solid B

Here is some information about solid A and solid B

|                           | solid A | solid B   |
|---------------------------|---------|-----------|
| Height (cm)               | $3^x$   |           |
| Area (cm <sup>2</sup> )   | 7776    | 486       |
| Volume (cm <sup>3</sup> ) | $8^x$   | $2^{x+4}$ |

Work out the height of solid B

Give your answer as a decimal.

\_\_\_\_\_

\_\_\_\_\_

..... cm

(Total for Question 23 is 5 marks)

\_\_\_\_\_

## PAST PAPER SOLUTION

Q#: 23

Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 23****Marks: 5****TOPIC: Area & Volume of Similar Shapes**

Nov 2023 P1H - Nov 2023 | P1H

**23 Solid A is similar to solid B**Here is some information about solid **A** and solid **B**

|                           | solid <b>A</b> | solid <b>B</b> |
|---------------------------|----------------|----------------|
| Height (cm)               | $3^x$          |                |
| Area (cm <sup>2</sup> )   | 7776           | 486            |
| Volume (cm <sup>3</sup> ) | $8^x$          | $2^{x+4}$      |

Work out the height of solid **B**

Give your answer as a decimal.

---

---

..... cm

---

**(Total for Question 23 is 5 marks)**

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## PAST PAPER SOLUTION

Q#: 23

Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 24****Marks: 4****TOPIC: Transformations of Graphs**

Nov 2023 P1H - Nov 2023 | P1H

**24** The curve with equation  $f(x) = 5x^2 + 9x + 2$  is transformed to the curve with equation

$$g(x) = 5(x+4)^2 + 9(x+4) + 8 \text{ by the translation } \begin{pmatrix} a \\ b \end{pmatrix}$$

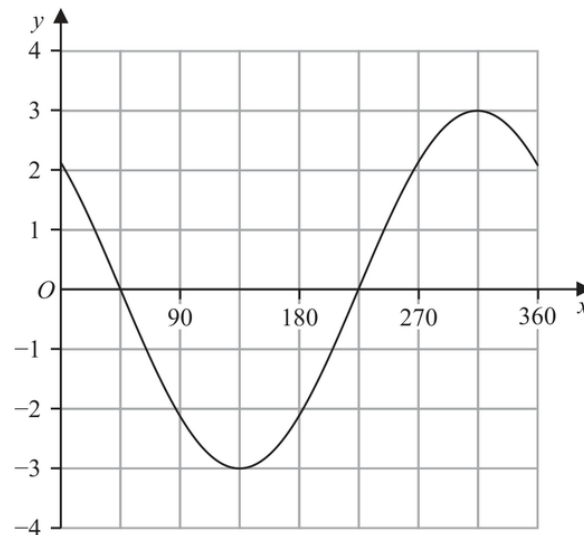
(a) Write down the value of  $a$  and the value of  $b$

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

(2)

The graph of  $y = p \cos(x+q)^\circ$  for  $0 \leq x \leq 360$  is drawn on the grid below.



Given that  $p > 0$  and  $0 < q < 360$

(b) find the value of  $p$  and the value of  $q$

$$p = \dots\dots\dots$$

$$q = \dots\dots\dots$$

(2)

**(Total for Question 24 is 4 marks)**

## PAST PAPER SOLUTION

Q#: 24

Marks: 4

TOPIC: **Transformations of Graphs**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 24****Marks: 4****TOPIC: Transformations of Graphs**

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**24** The curve with equation  $f(x) = 5x^2 + 9x + 2$  is transformed to the curve with equation

$$g(x) = 5(x+4)^2 + 9(x+4) + 8 \text{ by the translation } \begin{pmatrix} a \\ b \end{pmatrix}$$

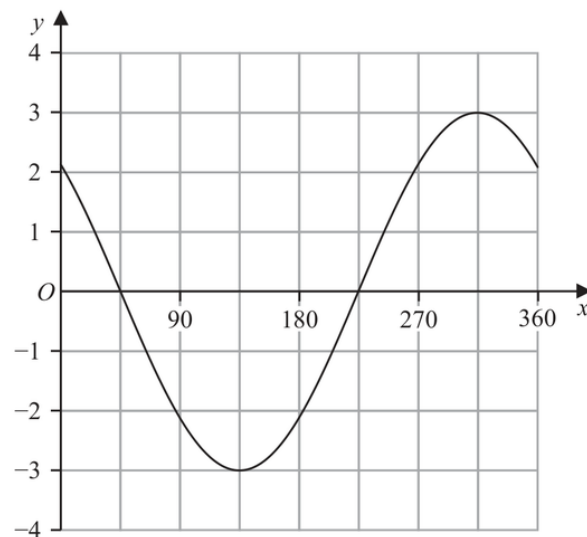
(a) Write down the value of  $a$  and the value of  $b$

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

(2)

The graph of  $y = p \cos(x+q)^\circ$  for  $0 \leq x \leq 360$  is drawn on the grid below.



Given that  $p > 0$  and  $0 < q < 360$

(b) find the value of  $p$  and the value of  $q$

$$p = \dots\dots\dots$$

$$q = \dots\dots\dots$$

(2)

**(Total for Question 24 is 4 marks)**

## PAST PAPER SOLUTION

Q#: 24

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TOPIC: **Transformations of Graphs**

Nov 2023 P1H - Nov 2023 | P1H

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA